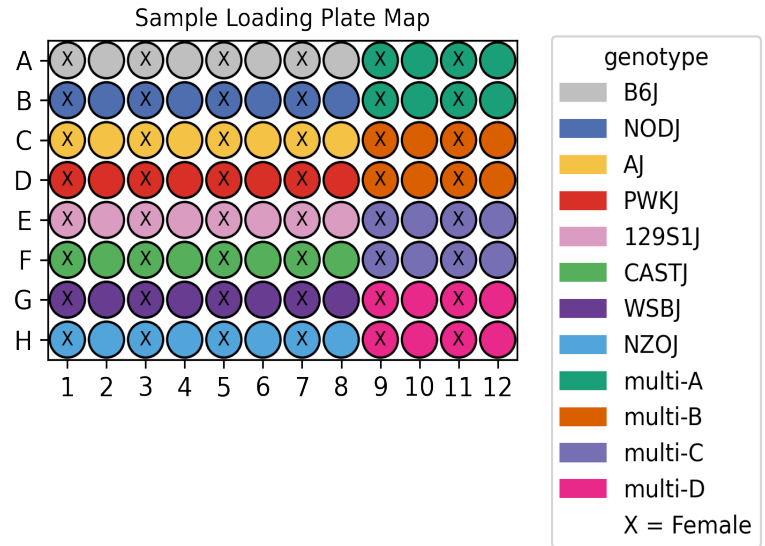


# UCI/Caltech IGVF Dataset Pipeline Report

|                       | igvf_010                                                     |
|-----------------------|--------------------------------------------------------------|
| Kit                   | WT_mega                                                      |
| Chemistry             | Parse v2                                                     |
| Sample type           | nuclei                                                       |
| Genotypes             | B6J<br>NODJ<br>AJ<br>PWKJ<br>129S1J<br>CASTJ<br>WSBJ<br>NZOJ |
| Tissues               | Kidney<br>Gastrocnemius                                      |
| Subpool               | Subpool_4                                                    |
| Measurment Set        | IGVFDS6137XZMZ                                               |
| Exome capture subpool | No                                                           |
| Expected nuclei       | 67,000                                                       |



## 1. Sample multiplexing

Kidney was the main tissue on the plate while Gastrocnemius was multiplexed.

Table 1: Genotype multiplexing key

| Pair    | Multiplexed Genotype 1 | Multiplexed Genotype 2 |
|---------|------------------------|------------------------|
| multi-A | B6J                    | NODJ                   |
| multi-B | AJ                     | PWKJ                   |
| multi-C | 129S1J                 | CASTJ                  |
| multi-D | NZOJ                   | WSBJ                   |

## 2. Alignment

A total of 1,612,033,505 reads were aligned using GENCODE vM32 with mm39 assembly.

Table 2: Pseudoalignment stats

| Subpool   | Total reads   | Total UMIs  | Fraction duplicated UMIs | Total pseudoaligned | Pct. pseudoaligned |
|-----------|---------------|-------------|--------------------------|---------------------|--------------------|
| Subpool_4 | 1,612,033,505 | 675,704,950 | 0.422                    | 1,168,793,864       | 72.5               |

### 3. Cell recovery

A total of 251,149 nuclei in Subpool\_4 have  $\geq 200$  UMIs and 80,701 nuclei have  $\geq 500$  UMIs.

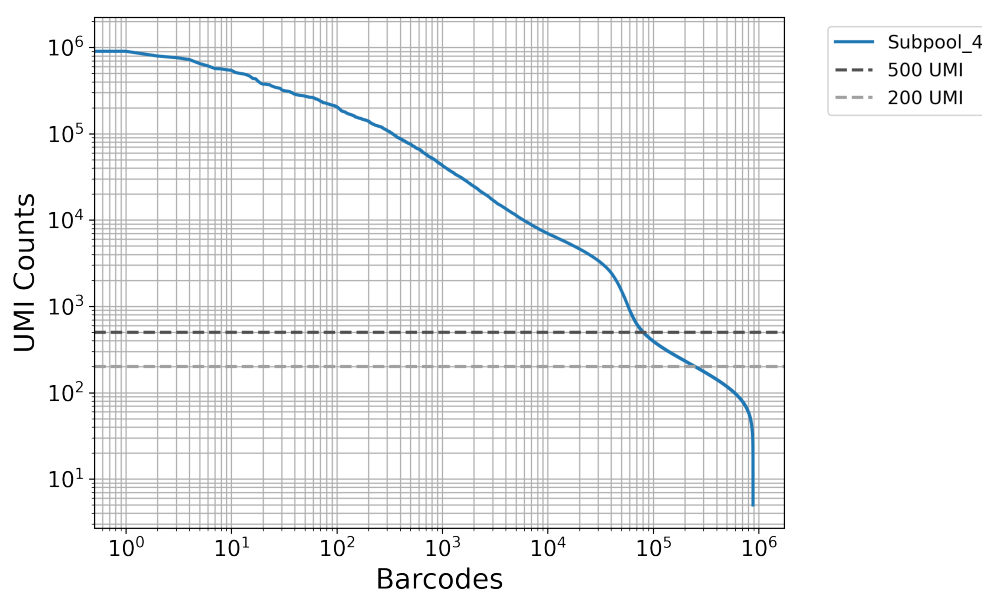


Table 3: QC stats for nuclei  $\geq 200$  UMI

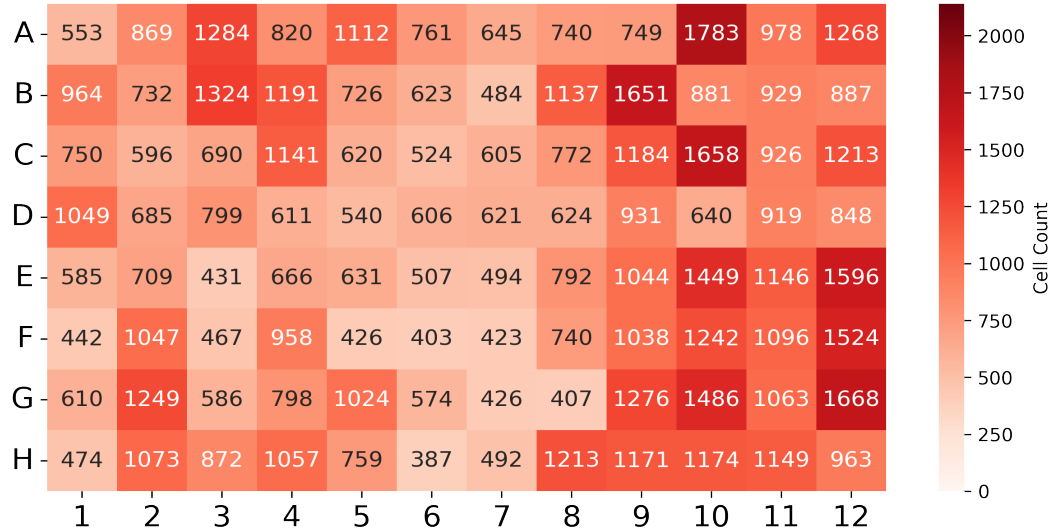
| Subpool   | Number of nuclei | Median UMIs | Mean UMIs | Median genes | Mean genes |
|-----------|------------------|-------------|-----------|--------------|------------|
| Subpool_4 | 251,149          | 326         | 1,790.9   | 263          | 684.7      |

Table 4: QC stats for nuclei  $\geq 500$  UMI

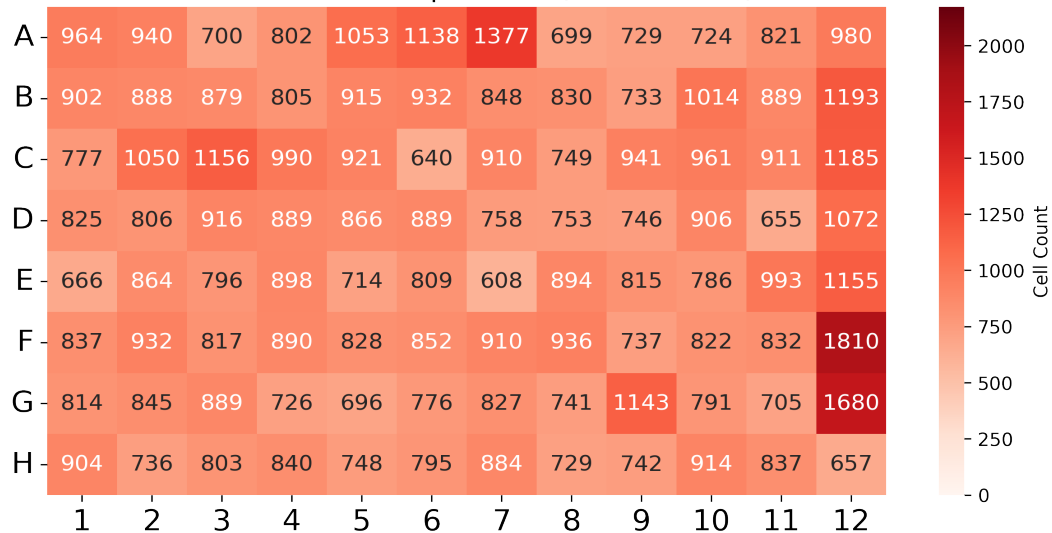
| Subpool   | Number of nuclei | Median UMIs | Mean UMIs | Median genes | Mean genes |
|-----------|------------------|-------------|-----------|--------------|------------|
| Subpool_4 | 80,701           | 2,407       | 4,967.2   | 1,241        | 1,635.6    |

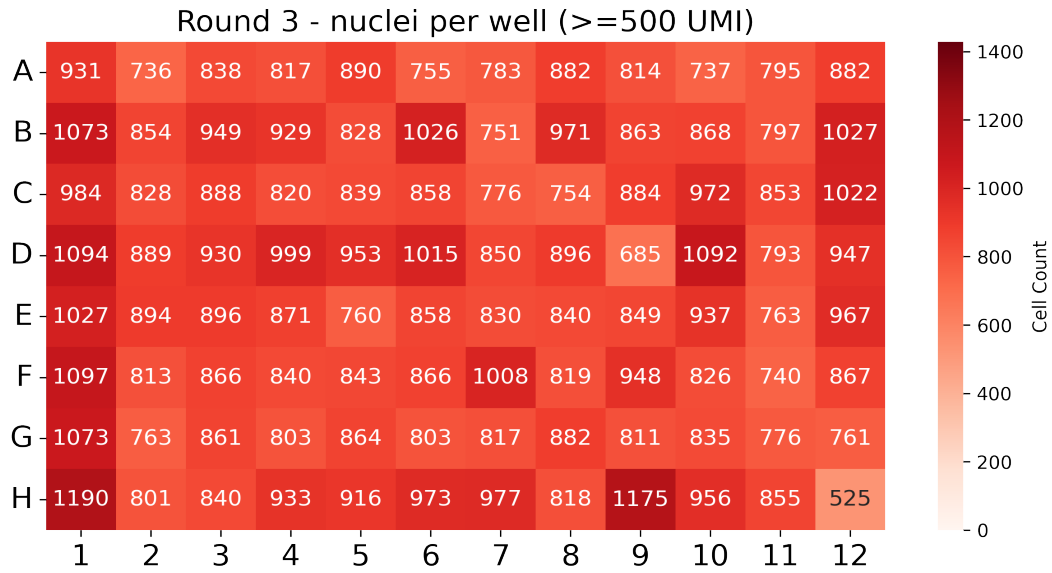
In this combinatorial barcoding assay, nuclei are barcoded in 96-well plates over three rounds. The first round assigns sample-specific barcodes, while the next two are applied to the pooled mix. Even distribution in the first round is important to ensure consistent sample representation.

Round 1 - nuclei per well ( $\geq 500$  UMI)



Round 2 - nuclei per well ( $\geq 500$  UMI)





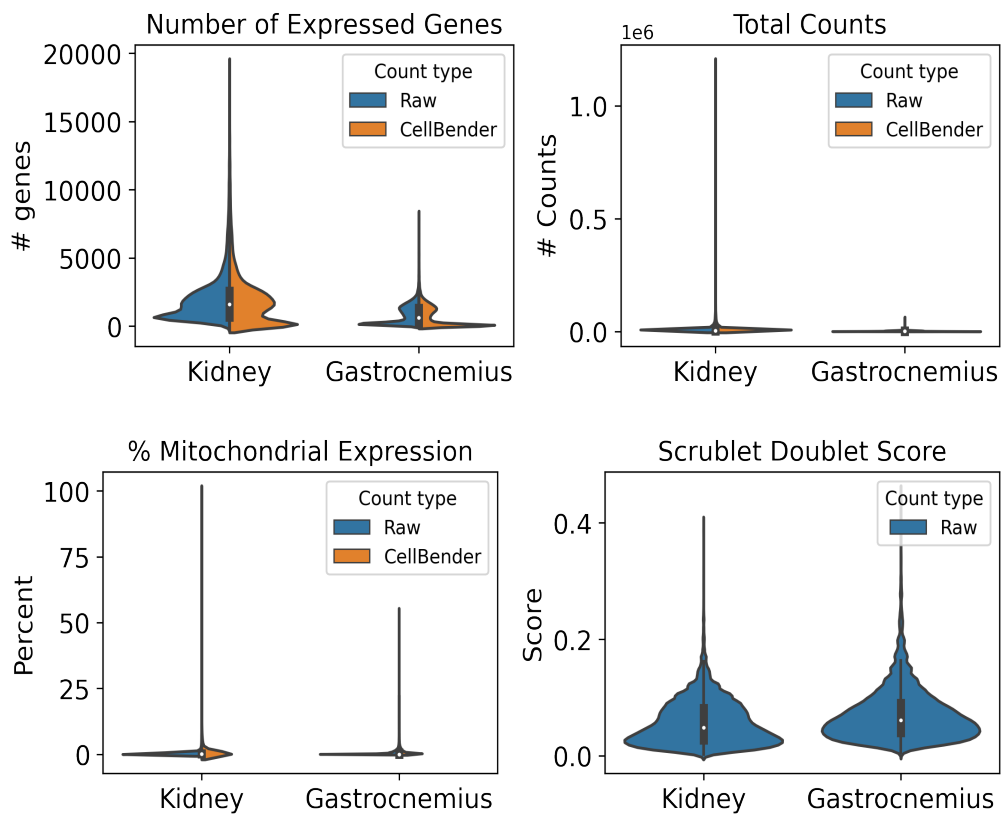
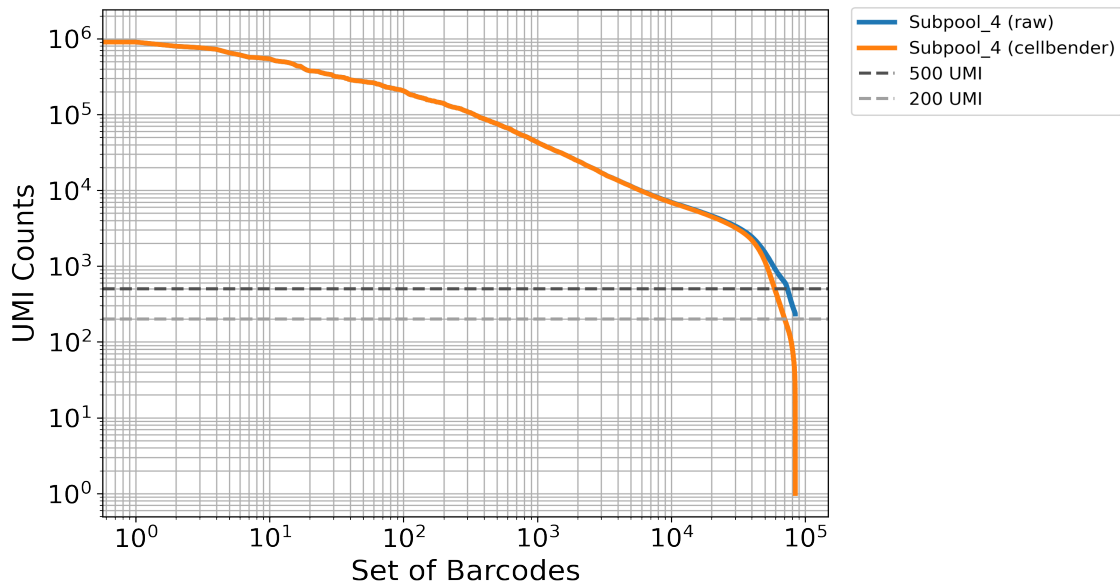
## 4. Background removal

Table 5: Cellbender parameters

| Subpool   | droplets_included | learning_rate | expected_cells |
|-----------|-------------------|---------------|----------------|
| Subpool_4 | 200000            | 5e-05         | 67000          |

Table 6: Cellbender stats

| Subpool   | Percent counts removed | Found nuclei | Mean denoised counts |
|-----------|------------------------|--------------|----------------------|
| Subpool_4 | 5.2                    | 84,450       | 4,496.5              |



## 5. Barcode sample map

The first round corresponds to sample barcoding. In Parse Bio assays, two barcodes are included in the first round: a random hexamer (R) and oligo-dT (T) barcode per well (parse barcode type). This reduces 3-prime bias and allows for the capture of

full-length transcripts. These sample level barcodes can be found in the barcode-1 region in the seqspec, and begin at position 16 (0-based) in the full 24-nt cell barcode. For the WT\_mega kit, samples are distributed across 96 wells in round 1. The following table includes our lab sample ID as sample descriptions and IGVF sample accessions.

Table 7: Barcode sample map

| barcode  | sample accession | sample description | position | parse barcode type | well |
|----------|------------------|--------------------|----------|--------------------|------|
| CATCATCC | IGVFSM0349VWWY   | 016_B6J_10F_09     | 16       | R                  | A1   |
| CATTCCTA | IGVFSM0349VWWY   | 016_B6J_10F_09     | 16       | T                  | A1   |
| CTGCTTTG | IGVFSM3241ASWH   | 017_B6J_10M_09     | 16       | R                  | A2   |
| CTTCATCA | IGVFSM3241ASWH   | 017_B6J_10M_09     | 16       | T                  | A2   |
| CTAAGGGA | IGVFSM9474HYEG   | 018_B6J_10F_09     | 16       | R                  | A3   |
| CCTATATC | IGVFSM9474HYEG   | 018_B6J_10F_09     | 16       | T                  | A3   |
| GCTTATAG | IGVFSM9788LCSW   | 019_B6J_10M_09     | 16       | R                  | A4   |
| ACATTTAC | IGVFSM9788LCSW   | 019_B6J_10M_09     | 16       | T                  | A4   |
| TCTGATCC | IGVFSM4800CIZG   | 020_B6J_10F_09     | 16       | R                  | A5   |
| ACTTAGCT | IGVFSM4800CIZG   | 020_B6J_10F_09     | 16       | T                  | A5   |
| TCTCTTGG | IGVFSM7243TNKF   | 021_B6J_10M_09     | 16       | R                  | A6   |
| CCAATTCT | IGVFSM7243TNKF   | 021_B6J_10M_09     | 16       | T                  | A6   |
| CAATTTCC | IGVFSM8674PTVY   | 024_B6J_10F_09     | 16       | R                  | A7   |
| GCCTATCT | IGVFSM8674PTVY   | 024_B6J_10F_09     | 16       | T                  | A7   |
| AGTCTCTT | IGVFSM3060AXDG   | 025_B6J_10M_09     | 16       | R                  | A8   |
| ATGCTGCT | IGVFSM3060AXDG   | 025_B6J_10M_09     | 16       | T                  | A8   |
| TGCTGCTC | IGVFSM4575TTYT   | 016_B6J_10F_16     | 16       | R                  | A9   |
| CATTTACA | IGVFSM4575TTYT   | 016_B6J_10F_16     | 16       | T                  | A9   |
| TGCTGCTC | IGVFSM4279QZLF   | 066_NODJ_10F_16    | 16       | R                  | A9   |
| CATTTACA | IGVFSM4279QZLF   | 066_NODJ_10F_16    | 16       | T                  | A9   |
| GTATTTCC | IGVFSM2276CJGW   | 017_B6J_10M_16     | 16       | R                  | A10  |
| ACTCGTAA | IGVFSM2276CJGW   | 017_B6J_10M_16     | 16       | T                  | A10  |
| GTATTTCC | IGVFSM1252BKWG   | 067_NODJ_10M_16    | 16       | R                  | A10  |
| ACTCGTAA | IGVFSM1252BKWG   | 067_NODJ_10M_16    | 16       | T                  | A10  |
| TTCCTGTG | IGVFSM0465UALO   | 018_B6J_10F_16     | 16       | R                  | A11  |
| CCTTTGCA | IGVFSM0465UALO   | 018_B6J_10F_16     | 16       | T                  | A11  |
| TTCCTGTG | IGVFSM8410WIND   | 074_NODJ_10F_16    | 16       | R                  | A11  |

|          |                |                 |    |   |     |
|----------|----------------|-----------------|----|---|-----|
| CCTTTGCA | IGVFSM8410WIND | 074_NODJ_10F_16 | 16 | T | A11 |
| GCTGCTTC | IGVFSM3666GJLM | 019_B6J_10M_16  | 16 | R | A12 |
| ACTCCTGC | IGVFSM3666GJLM | 019_B6J_10M_16  | 16 | T | A12 |
| GCTGCTTC | IGVFSM7620ICFV | 069_NODJ_10M_16 | 16 | R | A12 |
| ACTCCTGC | IGVFSM7620ICFV | 069_NODJ_10M_16 | 16 | T | A12 |
| TATGTGTC | IGVFSM7878IZMN | 066_NODJ_10F_09 | 16 | R | B1  |
| ATTTGGCA | IGVFSM7878IZMN | 066_NODJ_10F_09 | 16 | T | B1  |
| CAATTCTC | IGVFSM4726ZWYC | 067_NODJ_10M_09 | 16 | R | B2  |
| TTATTCTG | IGVFSM4726ZWYC | 067_NODJ_10M_09 | 16 | T | B2  |
| TGGTCTCC | IGVFSM4214PFWU | 068_NODJ_10F_09 | 16 | R | B3  |
| TCATGCTC | IGVFSM4214PFWU | 068_NODJ_10F_09 | 16 | T | B3  |
| GCTCTTTA | IGVFSM8898IMBT | 069_NODJ_10M_09 | 16 | R | B4  |
| CATACTTC | IGVFSM8898IMBT | 069_NODJ_10M_09 | 16 | T | B4  |
| GCTGCATG | IGVFSM7567RTBV | 070_NODJ_10F_09 | 16 | R | B5  |
| CCGTTCTA | IGVFSM7567RTBV | 070_NODJ_10F_09 | 16 | T | B5  |
| ACTCATTT | IGVFSM4531FRYY | 071_NODJ_10M_09 | 16 | R | B6  |
| GCTTCATA | IGVFSM4531FRYY | 071_NODJ_10M_09 | 16 | T | B6  |
| AGTCTTGG | IGVFSM4518CECN | 072_NODJ_10F_09 | 16 | R | B7  |
| CTCTGTGC | IGVFSM4518CECN | 072_NODJ_10F_09 | 16 | T | B7  |
| GGTTCTTC | IGVFSM4828VZOR | 073_NODJ_10M_09 | 16 | R | B8  |
| CCCTTATA | IGVFSM4828VZOR | 073_NODJ_10M_09 | 16 | T | B8  |
| TCATGTTG | IGVFSM4471UQDS | 020_B6J_10F_16  | 16 | R | B9  |
| ACTGCTCT | IGVFSM4471UQDS | 020_B6J_10F_16  | 16 | T | B9  |
| TCATGTTG | IGVFSM3414NFLU | 070_NODJ_10F_16 | 16 | R | B9  |
| ACTGCTCT | IGVFSM3414NFLU | 070_NODJ_10F_16 | 16 | T | B9  |
| ATTTTGCC | IGVFSM0131WZLT | 021_B6J_10M_16  | 16 | R | B10 |
| CTCTAATC | IGVFSM0131WZLT | 021_B6J_10M_16  | 16 | T | B10 |
| ATTTTGCC | IGVFSM9301WUUD | 071_NODJ_10M_16 | 16 | R | B10 |
| CTCTAATC | IGVFSM9301WUUD | 071_NODJ_10M_16 | 16 | T | B10 |
| CTTCTGTA | IGVFSM5168ZHLV | 024_B6J_10F_16  | 16 | R | B11 |
| ACCCTTGC | IGVFSM5168ZHLV | 024_B6J_10F_16  | 16 | T | B11 |
| CTTCTGTA | IGVFSM5744JVUK | 072_NODJ_10F_16 | 16 | R | B11 |
| ACCCTTGC | IGVFSM5744JVUK | 072_NODJ_10F_16 | 16 | T | B11 |
| GTCCATCT | IGVFSM2944JZTR | 025_B6J_10M_16  | 16 | R | B12 |
| ATCTTAGG | IGVFSM2944JZTR | 025_B6J_10M_16  | 16 | T | B12 |

|          |                |                 |    |   |     |
|----------|----------------|-----------------|----|---|-----|
| GTCCATCT | IGVFSM6017WKVH | 073_NODJ_10M_16 | 16 | R | B12 |
| ATCTTAGG | IGVFSM6017WKVH | 073_NODJ_10M_16 | 16 | T | B12 |
| GCTATCTC | IGVFSM3057IFAI | 026_AJ_10F_09   | 16 | R | C1  |
| CATGTCTC | IGVFSM3057IFAI | 026_AJ_10F_09   | 16 | T | C1  |
| TAGTTTCC | IGVFSM8586FNQO | 029_AJ_10M_09   | 16 | R | C2  |
| TCATTGCA | IGVFSM8586FNQO | 029_AJ_10M_09   | 16 | T | C2  |
| TCCATTAT | IGVFSM3536NVMG | 028_AJ_10F_09   | 16 | R | C3  |
| ACACCTTT | IGVFSM3536NVMG | 028_AJ_10F_09   | 16 | T | C3  |
| AGGATTAA | IGVFSM4523NJGP | 031_AJ_10M_09   | 16 | R | C4  |
| AATTTCTC | IGVFSM4523NJGP | 031_AJ_10M_09   | 16 | T | C4  |
| AATCTTTC | IGVFSM7352LZKM | 030_AJ_10F_09   | 16 | R | C5  |
| ATTCATGG | IGVFSM7352LZKM | 030_AJ_10F_09   | 16 | T | C5  |
| GTCATATG | IGVFSM6020DPIE | 033_AJ_10M_09   | 16 | R | C6  |
| ACTTTACC | IGVFSM6020DPIE | 033_AJ_10M_09   | 16 | T | C6  |
| GTGCTTCC | IGVFSM3951CGSL | 032_AJ_10F_09   | 16 | R | C7  |
| CTTCTAAC | IGVFSM3951CGSL | 032_AJ_10F_09   | 16 | T | C7  |
| ATGTGTTG | IGVFSM3981QKEE | 035_AJ_10M_09   | 16 | R | C8  |
| CTATTTCA | IGVFSM3981QKEE | 035_AJ_10M_09   | 16 | T | C8  |
| CCATCTTG | IGVFSM8801AFPS | 026_AJ_10F_16   | 16 | R | C9  |
| TCTCATGC | IGVFSM8801AFPS | 026_AJ_10F_16   | 16 | T | C9  |
| CCATCTTG | IGVFSM2436WWNM | 076_PWKJ_10F_16 | 16 | R | C9  |
| TCTCATGC | IGVFSM2436WWNM | 076_PWKJ_10F_16 | 16 | T | C9  |
| TACTGTCT | IGVFSM8141CEOO | 027_AJ_10M_16   | 16 | R | C10 |
| ATCCTTAC | IGVFSM8141CEOO | 027_AJ_10M_16   | 16 | T | C10 |
| TACTGTCT | IGVFSM8807FWBB | 077_PWKJ_10M_16 | 16 | R | C10 |
| ATCCTTAC | IGVFSM8807FWBB | 077_PWKJ_10M_16 | 16 | T | C10 |
| TTCATCGC | IGVFSM4118LWUK | 028_AJ_10F_16   | 16 | R | C11 |
| TAAATATC | IGVFSM4118LWUK | 028_AJ_10F_16   | 16 | T | C11 |
| TTCATCGC | IGVFSM9269MUDI | 078_PWKJ_10F_16 | 16 | R | C11 |
| TAAATATC | IGVFSM9269MUDI | 078_PWKJ_10F_16 | 16 | T | C11 |
| ACTGTGGG | IGVFSM2182IYXX | 031_AJ_10M_16   | 16 | R | C12 |
| TTACCTGC | IGVFSM2182IYXX | 031_AJ_10M_16   | 16 | T | C12 |
| ACTGTGGG | IGVFSM2389EAOO | 079_PWKJ_10M_16 | 16 | R | C12 |
| TTACCTGC | IGVFSM2389EAOO | 079_PWKJ_10M_16 | 16 | T | C12 |
| TCTGTGCC | IGVFSM9088FTIY | 076_PWKJ_10F_09 | 16 | R | D1  |



|          |                |                   |    |   |     |
|----------|----------------|-------------------|----|---|-----|
| CACTTTCA | IGVFSM9088FTIY | 076_PWKJ_10F_09   | 16 | T | D1  |
| TCAATCTC | IGVFSM2207LORL | 077_PWKJ_10M_09   | 16 | R | D2  |
| CACCTTTA | IGVFSM2207LORL | 077_PWKJ_10M_09   | 16 | T | D2  |
| GTCCTCTG | IGVFSM4211TCCR | 078_PWKJ_10F_09   | 16 | R | D3  |
| CTGACTTC | IGVFSM4211TCCR | 078_PWKJ_10F_09   | 16 | T | D3  |
| TTACATTC | IGVFSM4917CZZF | 079_PWKJ_10M_09   | 16 | R | D4  |
| CATTTGGA | IGVFSM4917CZZF | 079_PWKJ_10M_09   | 16 | T | D4  |
| ATTCTGTC | IGVFSM4983BATQ | 080_PWKJ_10F_09   | 16 | R | D5  |
| GCTCTACT | IGVFSM4983BATQ | 080_PWKJ_10F_09   | 16 | T | D5  |
| TGTGTATG | IGVFSM8581GLVP | 081_PWKJ_10M_09   | 16 | R | D6  |
| GTTACGTA | IGVFSM8581GLVP | 081_PWKJ_10M_09   | 16 | T | D6  |
| TCCATTTG | IGVFSM3357EWRJ | 084_PWKJ_10F_09   | 16 | R | D7  |
| CCTGTTGC | IGVFSM3357EWRJ | 084_PWKJ_10F_09   | 16 | T | D7  |
| TTAGCTTC | IGVFSM8335WDIG | 085_PWKJ_10M_09   | 16 | R | D8  |
| CTATCATC | IGVFSM8335WDIG | 085_PWKJ_10M_09   | 16 | T | D8  |
| GTGCTTGA | IGVFSM5510EXRQ | 030_AJ_10F_16     | 16 | R | D9  |
| GCTATCAT | IGVFSM5510EXRQ | 030_AJ_10F_16     | 16 | T | D9  |
| GTGCTTGA | IGVFSM0334GXDS | 080_PWKJ_10F_16   | 16 | R | D9  |
| GCTATCAT | IGVFSM0334GXDS | 080_PWKJ_10F_16   | 16 | T | D9  |
| GTTTGTGA | IGVFSM4001HPLV | 033_AJ_10M_16     | 16 | R | D10 |
| ACATTCAT | IGVFSM4001HPLV | 033_AJ_10M_16     | 16 | T | D10 |
| GTTTGTGA | IGVFSM4237VCZG | 081_PWKJ_10M_16   | 16 | R | D10 |
| ACATTCAT | IGVFSM4237VCZG | 081_PWKJ_10M_16   | 16 | T | D10 |
| GAAATTAG | IGVFSM2425SGDB | 034_AJ_10F_16     | 16 | R | D11 |
| TTCGCTAC | IGVFSM2425SGDB | 034_AJ_10F_16     | 16 | T | D11 |
| GAAATTAG | IGVFSM1365IGJT | 084_PWKJ_10F_16   | 16 | R | D11 |
| TTCGCTAC | IGVFSM1365IGJT | 084_PWKJ_10F_16   | 16 | T | D11 |
| GCAAATTC | IGVFSM7990NFIW | 035_AJ_10M_16     | 16 | R | D12 |
| CATTCTAC | IGVFSM7990NFIW | 035_AJ_10M_16     | 16 | T | D12 |
| GCAAATTC | IGVFSM1986OPVB | 085_PWKJ_10M_16   | 16 | R | D12 |
| CATTCTAC | IGVFSM1986OPVB | 085_PWKJ_10M_16   | 16 | T | D12 |
| GAGGTTGA | IGVFSM7015STPL | 036_129S1J_10F_09 | 16 | R | E1  |
| CACTTATC | IGVFSM7015STPL | 036_129S1J_10F_09 | 16 | T | E1  |
| CCTGTCTG | IGVFSM3870IKYE | 037_129S1J_10M_09 | 16 | R | E2  |
| ATAAGCTC | IGVFSM3870IKYE | 037_129S1J_10M_09 | 16 | T | E2  |

|          |                |                   |    |   |     |
|----------|----------------|-------------------|----|---|-----|
| GTGGGTTC | IGVFSM2985VTWQ | 038_129S1J_10F_09 | 16 | R | E3  |
| TCATCCTG | IGVFSM2985VTWQ | 038_129S1J_10F_09 | 16 | T | E3  |
| TTTGCATC | IGVFSM1728JVMJ | 041_129S1J_10M_09 | 16 | R | E4  |
| CCTGGTAT | IGVFSM1728JVMJ | 041_129S1J_10M_09 | 16 | T | E4  |
| AGGTAATA | IGVFSM7656VCGX | 040_129S1J_10F_09 | 16 | R | E5  |
| TGGTATAC | IGVFSM7656VCGX | 040_129S1J_10F_09 | 16 | T | E5  |
| GTGCCTTC | IGVFSM3274UVBQ | 043_129S1J_10M_09 | 16 | R | E6  |
| TTGGGAGA | IGVFSM3274UVBQ | 043_129S1J_10M_09 | 16 | T | E6  |
| ATGTTTCC | IGVFSM9083BJCW | 044_129S1J_10F_09 | 16 | R | E7  |
| ACTTCATC | IGVFSM9083BJCW | 044_129S1J_10F_09 | 16 | T | E7  |
| CTTAATTC | IGVFSM2309TUXE | 045_129S1J_10M_09 | 16 | R | E8  |
| TCTCTAGC | IGVFSM2309TUXE | 045_129S1J_10M_09 | 16 | T | E8  |
| TCTGGCTC | IGVFSM6869OUVF | 042_129S1J_10F_16 | 16 | R | E9  |
| ATGCCCTT | IGVFSM6869OUVF | 042_129S1J_10F_16 | 16 | T | E9  |
| TCTGGCTC | IGVFSM0725SKMI | 086_CASTJ_10F_16  | 16 | R | E9  |
| ATGCCCTT | IGVFSM0725SKMI | 086_CASTJ_10F_16  | 16 | T | E9  |
| CATCATTT | IGVFSM5187OEBL | 037_129S1J_10M_16 | 16 | R | E10 |
| CCCAATTT | IGVFSM5187OEBL | 037_129S1J_10M_16 | 16 | T | E10 |
| CATCATTT | IGVFSM8092ISYK | 087_CASTJ_10M_16  | 16 | R | E10 |
| CCCAATTT | IGVFSM8092ISYK | 087_CASTJ_10M_16  | 16 | T | E10 |
| GTTGTCTC | IGVFSM7768VTDI | 038_129S1J_10F_16 | 16 | R | E11 |
| ACTATATA | IGVFSM7768VTDI | 038_129S1J_10F_16 | 16 | T | E11 |
| GTTGTCTC | IGVFSM3158KFTN | 092_CASTJ_10F_16  | 16 | R | E11 |
| ACTATATA | IGVFSM3158KFTN | 092_CASTJ_10F_16  | 16 | T | E11 |
| ATCTTCTG | IGVFSM1059YXHB | 041_129S1J_10M_16 | 16 | R | E12 |
| CTCTATAC | IGVFSM1059YXHB | 041_129S1J_10M_16 | 16 | T | E12 |
| ATCTTCTG | IGVFSM2176DUOO | 089_CASTJ_10M_16  | 16 | R | E12 |
| CTCTATAC | IGVFSM2176DUOO | 089_CASTJ_10M_16  | 16 | T | E12 |
| TGTTTGCC | IGVFSM7915MIHZ | 086_CASTJ_10F_09  | 16 | R | F1  |
| CTGTCTCA | IGVFSM7915MIHZ | 086_CASTJ_10F_09  | 16 | T | F1  |
| TTCTGTCA | IGVFSM1924DVVF | 087_CASTJ_10M_09  | 16 | R | F2  |
| GACCTTTC | IGVFSM1924DVVF | 087_CASTJ_10M_09  | 16 | T | F2  |
| ACGGACTC | IGVFSM8639WEBM | 088_CASTJ_10F_09  | 16 | R | F3  |
| GATTTGGC | IGVFSM8639WEBM | 088_CASTJ_10F_09  | 16 | T | F3  |
| TTTGGTCA | IGVFSM0501BSTV | 089_CASTJ_10M_09  | 16 | R | F4  |

|          |                |                   |    |   |     |
|----------|----------------|-------------------|----|---|-----|
| CGTCTAGG | IGVFSM0501BSTV | 089_CASTJ_10M_09  | 16 | T | F4  |
| TATCCGGG | IGVFSM6781WLWW | 090_CASTJ_10F_09  | 16 | R | F5  |
| TACTCGAA | IGVFSM6781WLWW | 090_CASTJ_10F_09  | 16 | T | F5  |
| TGTCATTC | IGVFSM7315OJAM | 091_CASTJ_10M_09  | 16 | R | F6  |
| CAGCCTTT | IGVFSM7315OJAM | 091_CASTJ_10M_09  | 16 | T | F6  |
| ATTCTCTG | IGVFSM3937HVNK | 094_CASTJ_10F_09  | 16 | R | F7  |
| CCTCATTA | IGVFSM3937HVNK | 094_CASTJ_10F_09  | 16 | T | F7  |
| TGGCTTCC | IGVFSM0131ZKOP | 093_CASTJ_10M_09  | 16 | R | F8  |
| CTTATACC | IGVFSM0131ZKOP | 093_CASTJ_10M_09  | 16 | T | F8  |
| TTGTTGCC | IGVFSM5139SKXC | 040_129S1J_10F_16 | 16 | R | F9  |
| TCTATTAC | IGVFSM5139SKXC | 040_129S1J_10F_16 | 16 | T | F9  |
| TTGTTGCC | IGVFSM7844SUGB | 090_CASTJ_10F_16  | 16 | R | F9  |
| TCTATTAC | IGVFSM7844SUGB | 090_CASTJ_10F_16  | 16 | T | F9  |
| GTCATCTC | IGVFSM7373DCLC | 043_129S1J_10M_16 | 16 | R | F10 |
| CCTGCATT | IGVFSM7373DCLC | 043_129S1J_10M_16 | 16 | T | F10 |
| GTCATCTC | IGVFSM1128BAML | 091_CASTJ_10M_16  | 16 | R | F10 |
| CCTGCATT | IGVFSM1128BAML | 091_CASTJ_10M_16  | 16 | T | F10 |
| TTGCTCAT | IGVFSM4259FCLE | 044_129S1J_10F_16 | 16 | R | F11 |
| CAATCCTT | IGVFSM4259FCLE | 044_129S1J_10F_16 | 16 | T | F11 |
| TTGCTCAT | IGVFSM4593QSDG | 094_CASTJ_10F_16  | 16 | R | F11 |
| CAATCCTT | IGVFSM4593QSDG | 094_CASTJ_10F_16  | 16 | T | F11 |
| CTGTCTGC | IGVFSM7248MKIA | 045_129S1J_10M_16 | 16 | R | F12 |
| TTGTCTTA | IGVFSM7248MKIA | 045_129S1J_10M_16 | 16 | T | F12 |
| CTGTCTGC | IGVFSM3110AEWG | 093_CASTJ_10M_16  | 16 | R | F12 |
| TTGTCTTA | IGVFSM3110AEWG | 093_CASTJ_10M_16  | 16 | T | F12 |
| TATATTCC | IGVFSM9077YRZW | 058_WSBJ_10F_09   | 16 | R | G1  |
| TCACTTTA | IGVFSM9077YRZW | 058_WSBJ_10F_09   | 16 | T | G1  |
| ATATTGGC | IGVFSM1334FZTF | 057_WSBJ_10M_09   | 16 | R | G2  |
| TGCTTGGG | IGVFSM1334FZTF | 057_WSBJ_10M_09   | 16 | T | G2  |
| GTGTCCTC | IGVFSM6194HVVO | 060_WSBJ_10F_09   | 16 | R | G3  |
| CGCTCATT | IGVFSM6194HVVO | 060_WSBJ_10F_09   | 16 | T | G3  |
| ATCTTCAT | IGVFSM7285SRRG | 059_WSBJ_10M_09   | 16 | R | G4  |
| GCCTCTAT | IGVFSM7285SRRG | 059_WSBJ_10M_09   | 16 | T | G4  |
| CGTGTTTG | IGVFSM2970OEMT | 062_WSBJ_10F_09   | 16 | R | G5  |
| GAGCACAA | IGVFSM2970OEMT | 062_WSBJ_10F_09   | 16 | T | G5  |

|          |                |                 |    |   |     |
|----------|----------------|-----------------|----|---|-----|
| TTGCATCC | IGVFSM0104SYAP | 061_WSBJ_10M_09 | 16 | R | G6  |
| CTCTTAAC | IGVFSM0104SYAP | 061_WSBJ_10M_09 | 16 | T | G6  |
| TCTTAATC | IGVFSM9060ZOSA | 096_WSBJ_10F_09 | 16 | R | G7  |
| TCTAGGCT | IGVFSM9060ZOSA | 096_WSBJ_10F_09 | 16 | T | G7  |
| TGCATTTT | IGVFSM4351BICU | 063_WSBJ_10M_09 | 16 | R | G8  |
| AATTCTGC | IGVFSM4351BICU | 063_WSBJ_10M_09 | 16 | T | G8  |
| GATGTTTC | IGVFSM5504RVNQ | 046_NZOJ_10F_16 | 16 | R | G9  |
| CATTCTCA | IGVFSM5504RVNQ | 046_NZOJ_10F_16 | 16 | T | G9  |
| GATGTTTC | IGVFSM0217YUAM | 058_WSBJ_10F_16 | 16 | R | G9  |
| CATTCTCA | IGVFSM0217YUAM | 058_WSBJ_10F_16 | 16 | T | G9  |
| ATCTTGTC | IGVFSM1584RGVM | 047_NZOJ_10M_16 | 16 | R | G10 |
| ACTTGCCT | IGVFSM1584RGVM | 047_NZOJ_10M_16 | 16 | T | G10 |
| ATCTTGTC | IGVFSM5571BGFV | 057_WSBJ_10M_16 | 16 | R | G10 |
| ACTTGCCT | IGVFSM5571BGFV | 057_WSBJ_10M_16 | 16 | T | G10 |
| TCATATTC | IGVFSM8444KPOJ | 048_NZOJ_10F_16 | 16 | R | G11 |
| ATCATTGC | IGVFSM8444KPOJ | 048_NZOJ_10F_16 | 16 | T | G11 |
| TCATATTC | IGVFSM0943LRWD | 064_WSBJ_10F_16 | 16 | R | G11 |
| ATCATTGC | IGVFSM0943LRWD | 064_WSBJ_10F_16 | 16 | T | G11 |
| TGGCCTCT | IGVFSM4489MUQD | 049_NZOJ_10M_16 | 16 | R | G12 |
| GTTCAACA | IGVFSM4489MUQD | 049_NZOJ_10M_16 | 16 | T | G12 |
| TGGCCTCT | IGVFSM8559RKMK | 059_WSBJ_10M_16 | 16 | R | G12 |
| GTTCAACA | IGVFSM8559RKMK | 059_WSBJ_10M_16 | 16 | T | G12 |
| CGTTGTCT | IGVFSM2869NKDL | 046_NZOJ_10F_09 | 16 | R | H1  |
| CCATTTGC | IGVFSM2869NKDL | 046_NZOJ_10F_09 | 16 | T | H1  |
| TCTTGTCA | IGVFSM8082YJJT | 047_NZOJ_10M_09 | 16 | R | H2  |
| GACTTTGC | IGVFSM8082YJJT | 047_NZOJ_10M_09 | 16 | T | H2  |
| TATTCCTG | IGVFSM4505WWVP | 048_NZOJ_10F_09 | 16 | R | H3  |
| ATTGGCTC | IGVFSM4505WWVP | 048_NZOJ_10F_09 | 16 | T | H3  |
| TCCATGTC | IGVFSM7894QTQT | 049_NZOJ_10M_09 | 16 | R | H4  |
| GTGCTAGC | IGVFSM7894QTQT | 049_NZOJ_10M_09 | 16 | T | H4  |
| TTGTCATC | IGVFSM9303HUHF | 050_NZOJ_10F_09 | 16 | R | H5  |
| CTTTCAAC | IGVFSM9303HUHF | 050_NZOJ_10F_09 | 16 | T | H5  |
| ATTTCTG  | IGVFSM0181XWPQ | 051_NZOJ_10M_09 | 16 | R | H6  |
| ACTATTGC | IGVFSM0181XWPQ | 051_NZOJ_10M_09 | 16 | T | H6  |
| GTGTCTCC | IGVFSM1587OINL | 052_NZOJ_10F_09 | 16 | R | H7  |

|          |                |                 |    |   |     |
|----------|----------------|-----------------|----|---|-----|
| ACTGGCTT | IGVFSM1587OINL | 052_NZOJ_10F_09 | 16 | T | H7  |
| GTGTGTGT | IGVFSM5770UUHX | 053_NZOJ_10M_09 | 16 | R | H8  |
| ATTAGGCT | IGVFSM5770UUHX | 053_NZOJ_10M_09 | 16 | T | H8  |
| TATGCTTC | IGVFSM5314ZJEY | 050_NZOJ_10F_16 | 16 | R | H9  |
| GCCTTTCA | IGVFSM5314ZJEY | 050_NZOJ_10F_16 | 16 | T | H9  |
| TATGCTTC | IGVFSM2242OSDW | 062_WSBJ_10F_16 | 16 | R | H9  |
| GCCTTTCA | IGVFSM2242OSDW | 062_WSBJ_10F_16 | 16 | T | H9  |
| ATGGTGTT | IGVFSM1469JLYZ | 051_NZOJ_10M_16 | 16 | R | H10 |
| ATTCTAGG | IGVFSM1469JLYZ | 051_NZOJ_10M_16 | 16 | T | H10 |
| ATGGTGTT | IGVFSM7709RATJ | 061_WSBJ_10M_16 | 16 | R | H10 |
| ATTCTAGG | IGVFSM7709RATJ | 061_WSBJ_10M_16 | 16 | T | H10 |
| GAATAATG | IGVFSM0037THNT | 052_NZOJ_10F_16 | 16 | R | H11 |
| CCTTACAT | IGVFSM0037THNT | 052_NZOJ_10F_16 | 16 | T | H11 |
| GAATAATG | IGVFSM4280ZZUH | 096_WSBJ_10F_16 | 16 | R | H11 |
| CCTTACAT | IGVFSM4280ZZUH | 096_WSBJ_10F_16 | 16 | T | H11 |
| CCTCTGTG | IGVFSM5658GGTC | 053_NZOJ_10M_16 | 16 | R | H12 |
| ACATTTGG | IGVFSM5658GGTC | 053_NZOJ_10M_16 | 16 | T | H12 |
| CCTCTGTG | IGVFSM5083IADN | 063_WSBJ_10M_16 | 16 | R | H12 |
| ACATTTGG | IGVFSM5083IADN | 063_WSBJ_10M_16 | 16 | T | H12 |

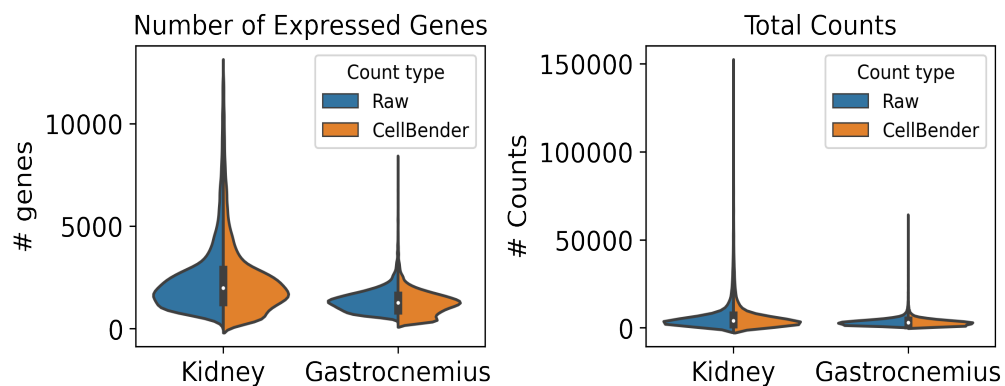
## 6. Description of data processing workflow and h5ad files

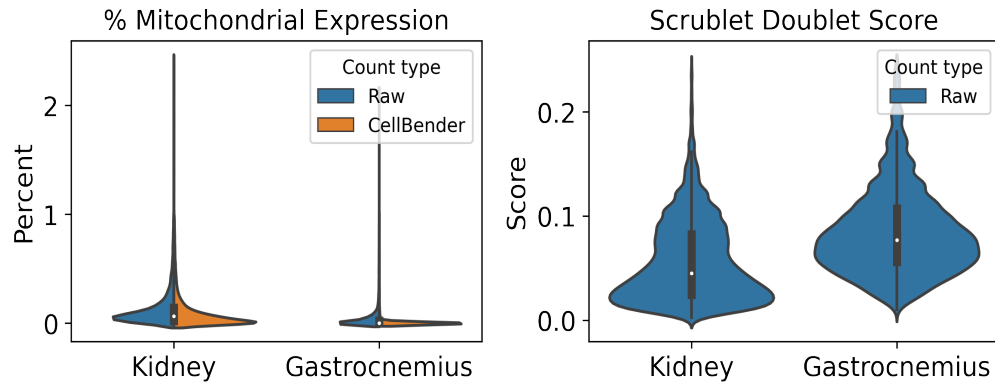
Data from Subpool\_4 were combined with all other subpools into one AnnData object per tissue as well as matching data from other plates for processing and cell type annotation. The AnnData contains detailed sample and cell metadata in the observation (obs) table and gene information in the variables table (var). Cell IDs (adata.obs.index) are re-formatted to be human-readable and unique across subpools and experiments by appending subpool and experiment IDs. The final AnnData contains both raw and CellBender denoised counts matrices in separate layers: adata.layers['raw\_counts'] and adata.layers['cellbender\_counts']. The current adata.X points to the CellBender denoised counts. Cells or nuclei are filtered by  $>0.5$  cell\_probability from CellBender analysis.

In addition to CellBender filtering, nuclei were also filtered by the following QC parameters:

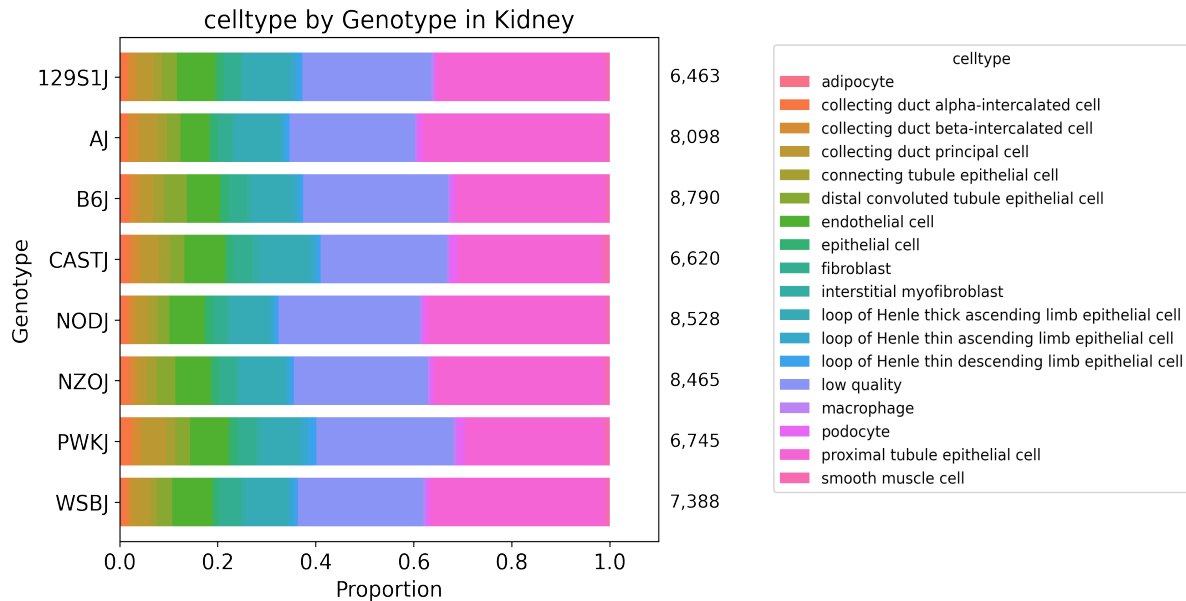
Table 8: QC thresholds

| Parameter             | Threshold |
|-----------------------|-----------|
| Minimum # UMIs        | 500.0     |
| Maximum # UMIs        | 150000.0  |
| Minimum # genes       | 250.0     |
| Maximum % mito.       | 1.0       |
| Maximum doublet score | 0.25      |





After filtering, data for each tissue was normalized and clustered using CellBender denoised counts. Normalization includes regression of technical parameters (number of genes expressed per cell and percent mitochondrial gene expression). Harmony was used to integrate exome capture and non-exome capture subpools for annotation. A Leiden clustering resolution of 1 was used for 54 total clusters in Kidney and 36 clusters in Gastrocnemius, using all tissue data across experiments. Cell type annotations were assigned per Leiden cluster or subcluster (leiden\_R) using expression of canonical cell type marker genes.



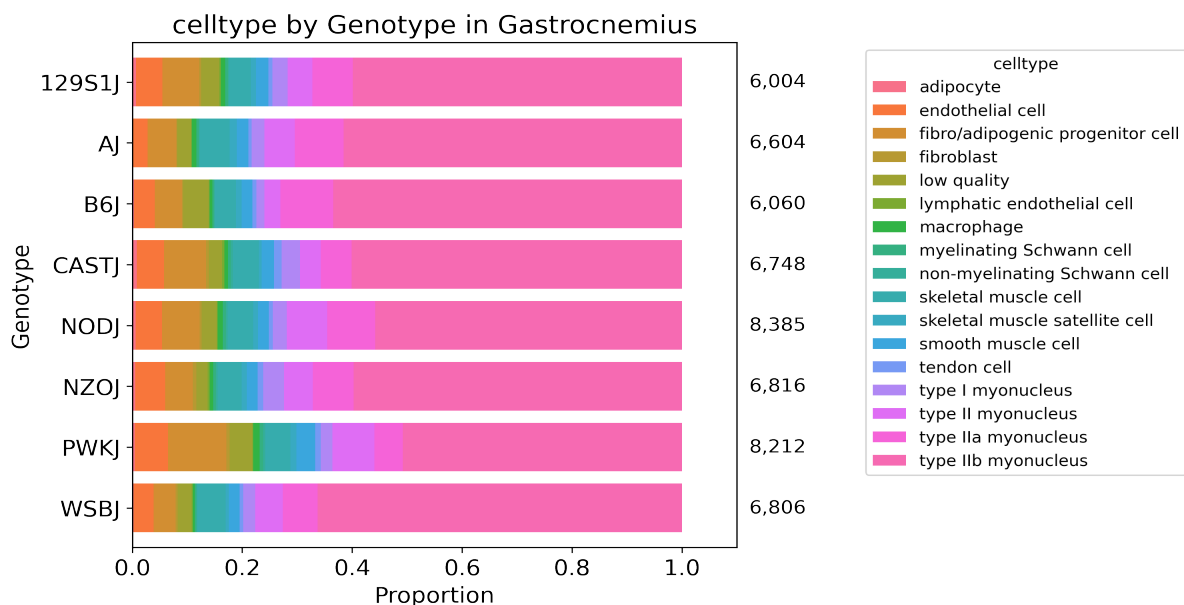


Table 9: Description of obs keys in h5ad file

| Key              | Description                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------|
| cellID           | Index of obs table. Includes barcoding wells for each round (1_2_3), subpool, and experiment. |
| lab_sample_id    | Human-readable sample name used in-house.                                                     |
| sample           | IGVF tissue accession.                                                                        |
| plate            | Experiment ID, e.g. igvf_003.                                                                 |
| subpool          | Subpool ID, e.g. Subpool_2. Number corresponds to Illumina index ID from Parse Bio kit.       |
| SampleType       | Nuclei or Cells.                                                                              |
| Tissue           | Human-readable tissue name, e.g. Kidney.                                                      |
| Sex              | Male or Female.                                                                               |
| Age              | Age in postnatal months (PNM) of the mouse, e.g. PNM_02.                                      |
| Genotype         | Mouse genotype/strain, e.g. B6J.                                                              |
| leiden           | Numeric Leiden cluster starting from 0.                                                       |
| leiden_R         | If necessary, Leiden clusters and any sub-clusters, e.g. 15_1.                                |
| subpool_type     | Exome capture (EX) or no exome capture (NO).                                                  |
| general_celltype | Broadest level of cell type annotations, e.g. neuron.                                         |
| general_CL_ID    | EMBL CL database ID for general cell type annotation, e.g. CL:0000540.                        |
| celltype         | Finest level of cell type annotation with a CL ID, e.g. Vip+ GABAergic cortical interneuron.  |
| CL_ID            | EMBL CL database ID for cell type annotation, CL:4023016                                      |
| subtype          | Finest level of cell type annotation that may not have a CL ID. Mostly matches celltype.      |
| Protocol         | Cell barcoding and library building protocol/kit, e.g. Parse_WT.                              |



|                       |                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------|
| Chemistry             | Protocol chemistry version, v2 or v3.                                                          |
| bc                    | 24-nucleotide sequence corresponding to 3 8-nucleotide barcode rounds (3, 2, 1).               |
| bc1_sequence          | 8-nucleotide sequence barcode for round 1.                                                     |
| bc2_sequence          | 8-nucleotide sequence barcode for round 2.                                                     |
| bc3_sequence          | 8-nucleotide sequence barcode for round 3.                                                     |
| bc1_well              | Round 1 well.                                                                                  |
| bc2_well              | Round 2 well.                                                                                  |
| bc3_well              | Round 2 well.                                                                                  |
| Row                   | Sample barcoding plate (round 1) row.                                                          |
| Column                | Sample barcoding plate (round 1) column.                                                       |
| well_type             | Single or Multiplexed.                                                                         |
| Mouse_Tissue_ID       | Same as lab_sample_id.                                                                         |
| Multiplexed_sample1   | Sample 1 from multiplexed well (if applicable)                                                 |
| Multiplexed_sample2   | Sample 2 from multiplexed well (if applicable)                                                 |
| DOB                   | Date of birth of the mouse in month/day/year format.                                           |
| Age_days              | Age of the mouse in days.                                                                      |
| Body_weight_g         | Body weight in grams of the mouse.                                                             |
| Estrus_cycle          | Estimated estrus stage of the mouse, if applicable.                                            |
| Dissection_date       | Date the tissue was harvested in month/day/year format.                                        |
| Dissection_time       | Time in hours:minutes (24-hour time) the mouse was sacrificed.                                 |
| ZT                    | Number of hours into the light cycle, from lights on (06:30, ZT0) to lights off (20:30, ZT14). |
| Dissector             | Initials of technician who dissected the tissue.                                               |
| Tissue_weight_mg      | Tissue weight in milligrams.                                                                   |
| Notes                 | Notes taken during sample collection, experiment, or data processing.                          |
| n_genes_by_counts_raw | The number of genes with at least 1 count, calculated using raw counts.                        |
| total_counts_raw      | Total counts (UMIs), calculated using raw counts.                                              |
| total_counts_mt_raw   | Total counts for mitochondrial genes, calculated using raw counts.                             |
| pct_counts_mt_raw     | Percent of total counts in mitochondrial genes, calculated using raw counts.                   |
| n_genes_by_counts_cb  | The number of genes with at least 1 count, calculated using CellBender counts.                 |
| total_counts_cb       | Total counts (UMIs), calculated using CellBender counts.                                       |
| total_counts_mt_cb    | Total counts for mitochondrial genes, calculated using CellBender counts.                      |
| pct_counts_mt_cb      | Percent of total counts in mitochondrial genes, calculated using CellBender counts.            |
| doublet_score         | Scrublet doublet score, calculated using CellBender counts.                                    |
| predicted_doublet     | Scrublet predicted doublet (True or False).                                                    |
| background_fraction   | Fraction of counts CellBender predicted to be background noise.                                |

|                    |                                                                         |
|--------------------|-------------------------------------------------------------------------|
| cell_probability   | CellBender probability that the cell is real.                           |
| cell_size          | CellBender size factor.                                                 |
| droplet_efficiency | CellBender statistic indicating transcript capture efficiency per cell. |
| B6J_klue_counts    | Estimated counts for B6J genotype from klue analysis.                   |
| NODJ_klue_counts   | Estimated counts for NODJ genotype from klue analysis.                  |
| AJ_klue_counts     | Estimated counts for AJ genotype from klue analysis.                    |
| PWKJ_klue_counts   | Estimated counts for PWKJ genotype from klue analysis.                  |
| 129S1J_klue_counts | Estimated counts for 129S1J genotype from klue analysis.                |
| CASTJ_klue_counts  | Estimated counts for CASTJ genotype from klue analysis.                 |
| WSBJ_klue_counts   | Estimated counts for WSBJ genotype from klue analysis.                  |
| NZOJ_klue_counts   | Estimated counts for NZOJ genotype from klue analysis.                  |

## 7. Package versions

Table 10: Package versions

| package    | version |
|------------|---------|
| kallisto   | 0.50.1  |
| bustools   | 0.43.2  |
| kb         | 0.28.2  |
| Python     | 3.9.0   |
| snakemake  | 7.32.0  |
| cellbender | 0.3.2   |
| scanpy     | 1.10.2  |
| scrublet   | 0.2.3   |
| anndata    | 0.10.8  |
| pandas     | 2.2.2   |
| numpy      | 1.26.4  |
| harmonypy  | 0.0.9   |

## 8. Contact information and GitHub

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Elisabeth Rebboah, [erebboah@uci.edu](mailto:erebboah@uci.edu)

Pipeline repo: [https://github.com/mortazavilab/parse\\_pipeline](https://github.com/mortazavilab/parse_pipeline)

Code used to make this report: [https://github.com/mortazavilab/8cube\\_manuscript/blob/main/plate\\_report/Subpool\\_report\\_igvf\\_010.ipynb](https://github.com/mortazavilab/8cube_manuscript/blob/main/plate_report/Subpool_report_igvf_010.ipynb)

Code used to process and annotate the data for main tissue: [https://github.com/mortazavilab/8cube\\_manuscript/blob/main/annotation/Kidney.ipynb](https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Kidney.ipynb)

Code used to process and annotate the data for multiplexed tissue: [https://github.com/mortazavilab/8cube\\_manuscript/blob/main/annotation/Gastrocnemius.ipynb](https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Gastrocnemius.ipynb)